

- JTAC capabilities (e.g., laser, infrared, video downlink).
- Factor ACM/FSCM.
- Radio calls required by controller.
- Intent for aircraft (e.g., CAS, multisensory imagery reconnaissance).
- ACM/FSCM/MCM.
- Provide a positive passing of the appropriate elements of brief, stack, mark, and control during BHO.
- CBRN targeting restrictions.
- Other remarks.

Table 84. Helicopter or Tiltrotor (LZ) Brief

LZ NAME or LOCATION:
 GRID/ELEVATION:
 SIZE (FEET):
 MARKED BY:

LANDING HEADING/DIRECTION:
 APPROACH FROM:
 LAND:
 DEPART TO:

FORMATION (TRAIL, ECHELON L/R, ONLINE, TRAIL):

WINDS (DIR/SPEED):

*Example extract table. This does not replace the use of the assault support landing table (ASLT) for USMC PZ operations.

CHALK 1	PAX #	DOOR LOAD	MARKING

CHALK 2	PAX #	DOOR LOAD	MARKING

HAZARDS:
 Obstructions:
 Terrain (brownout, pinnacle, etc.):
 Enemy (SALTA Report):

EXAMPLE CONTROLLER TO A/C PICK-UP CALLS:
 "Ready for extract"
 "Exfil, exfil, exfil"
 "CASEVAC, CASEVAC, CASEVAC"

EXAMPLE LZ BRIEF FROM CONTROLLER:
 "Metal 41, Broadsword 14 advise ready LZ brief."
 "LZ Napier, 200x200, marked by IR pointer, approach from south, land 330 echelon right, left door load. Winds N 5kts."
 "Chalk one 15 pax marked by IR strobe, chalk two 12 pax marked by IR buzzsaw."
 "Expect brownout, antenna 100m North of LZ, no known enemy activity."
 "Call ready for mark (smoke, rope, sparkle, etc.)."

Table 85. Battle Damage Estimate (BDE) Report

(aircraft call sign) (JTAC/FAC(A) call sign)
Type (1, 2, or 3) control, MOA (BOC or BOT), (ordnance requested)."

2. Friendly location/mark
"My position _____, marked by _____"
 (target reference point, grid, etc.) (beacon, IR strobe, etc.)
3. Target location
"Target location, _____"
 (magnetic bearing and range in meters, target reference point, grid, etc.)
4. Target description/mark
"_____, marked by _____."
 (target description) (IR, tracer, etc.)
5. Remarks and Restrictions (* items are restrictions):
- Laser-to-target line or pointer target line.
 - Desired type and number of ordnance or weapons effects (if not previously coordinated).
 - Surface-to-air threat, location, and type of SEAD.
 - Additional calls requested.
 - Additional remarks (gun-target line, weather, hazards).
 - *Final attack headings.
 - *Airspace coordination areas.
 - *Danger close and initials.
 - *Time on target/time to target.
 - *Post launch abort coordination and considerations.

Note: The rotary wing CAS 5-line should be passed as one transmission. If the restrictions portion is lengthy, it may be a separate transmission.

Legend:

BP—battle position
 BOC—bomb on coordinate
 BOT—bomb on target
 FAC(A)—forward air controller (airborne)

IR—infrared
 JTAC—joint terminal attack controller
 MOA—method of attack
 SEAD—suppression of enemy air defenses

Army Attack Aviation/SOF Rotary Wing Request and SOF Gunship CFF Format

1. Observer and Warning Order.
"_____, this is _____, fire mission, over."
 (aircraft call sign) (observer call sign)
2. Friendly Location and Mark.
"My position _____, marked by _____."
 (e.g., grid) (strobe, beacon, etc.)
3. Target Location.
"Target Location _____,"
 (bearing (magnetic) and range (meters), grid, etc.)
4. Target Description and Mark.
"_____, marked by _____."
 (target description) (infrared pointer, tracer, etc.)
5. Remarks: "

Table 33. Battle Damage Estimate (BDA) Report

Size: "_____"
 (number and type of equipment/personnel observed)
 Activity: "_____"
 (movement direction, stationary, dug-in)
 Location: "_____"
 Time: "_____"
 Remarks: "_____"
 (munitions expended, observed damage, mission number, and mission accomplished)
 Munitions Effective Assessment: "_____"
 Re-attack Recommendation: "_____"

_____, over."
 (clearance, danger close, at my command, threats, restrictions, etc.)

Notes:

1. Clearance. If airspace has been cleared between the employing aircraft and the target, transmission of this brief is clearance to fire unless "danger close", "at my command", or an additional method of control is stated.
2. Danger Close. The commander must accept responsibility for increased risk. State "cleared danger close" in line 5 (If required, provide "Danger Close" in line 1 for Army Attack Aviation/SOF RW) and pass the initials of the on-scene ground commander. This clearance may be preplanned.
3. At My Command. For positive control of the aircraft, state "at my command" on line 5 (If required, provide "At my command" in line 1 for Army Attack Aviation/SOF RW). The aircraft may call "ready", when ready. To command the aircraft to attack, the observer will say "(aircraft call sign) cleared to engage."
4. For "Multi-target" engagements: Provide lines 3 and 4 sequentially for each target prior to passing "Remarks" (If required, provide "Multi-target" in line 1 for Army Attack Aviation/SOF RW).
5. For synchronization of fires, methods of fire and control may be included in line 5. Refer to table 2, "Methods of Fire and Control", for additional measures.

Danger Close Limits

(Red Zones)

If the ordnance used is not in this list, DANGER CLOSE: 400 meters

Bombs		
Weapon	Guidance	DANGER CLOSE
CBU-87	N/A	245 meters
CBU-103	GPS	225 meters
GBU-12	Laser	270 meters
GBU-24	Laser	380 meters
GBU-31	GPS	465 meters
GBU-32	GPS	340 meters
GBU-38	GPS	270 meters
GBU-39	GPS/Laser	225 meters
GBU-49	GPS/Laser	565 meters
GBU-53	GPS/Laser/IR/Radar	185 meters
GBU-54	GPS/Laser	280 meters
MK-82	N/A	285 meters

Missiles & Rockets		
Weapon	Guidance	DANGER CLOSE
Hydra 70 APKWS	Laser	105 meters
Hydra 70	N/A	270 meters
Hydra 70 DAGR	N/A	105 meters
AGM-65D	IR	175 meters
AGM-65E2/L	Laser	
AGM-65K	IR	
AGM-114K	Laser	115 meters
AGM-114N	Laser	
AGM-114L	Radar	

Guns	
Weapon	DANGER CLOSE
30 mm HEDP	200 meters

SOF Gunship		7.62 Miniguns	100 meters
Weapon	DANGER CLOSE	20mm	150 meters
25mm GUA-12 Equalizer	60 meters		
40mm L60-Bofors Cannon	80 meters		
105mm M102 Howitzer	160 meters		

12 Steps of CAS

TFW 18E 12 Steps of CAS

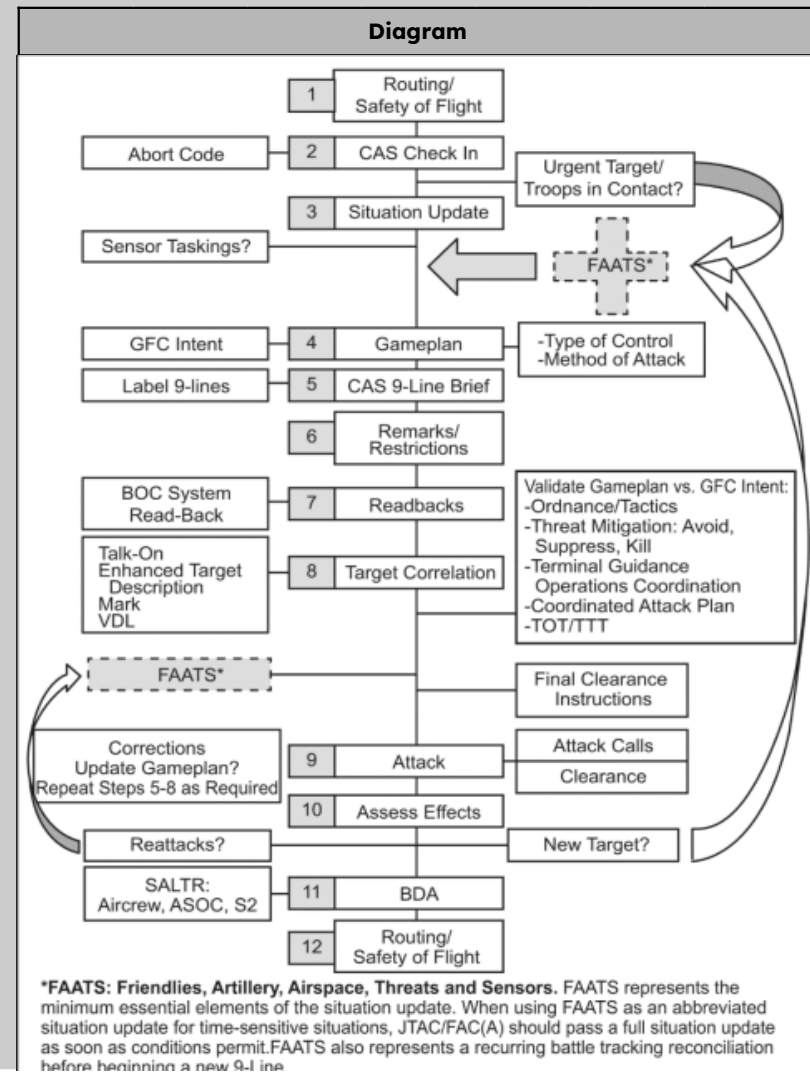
Explanation

1. Routing / Safety of Flight.
 - Three-dimensional directions from the current aircraft position to the intended holding point or area.
 - Holding point and altitude, once established, and which agency to contact.
 - Other aircraft on station.
 - Surface-to-air threats that may immediately affect close air support aircraft.
 - Any other safety of flight issues.
 - Pass any gun-target-line information (active or inactive) and azimuth/location.

2. CAS Check-in Briefing.
 - Aircrew: " _____, this is _____."
 - Mission number: (if required) " _____."
 - Number and type of aircraft: " _____."
 - Position and altitude: " _____."
 - Ordnance: " _____."
 - Playtime or time on station: " _____."
 - Capabilities: " _____"
 - (type of sensors, video downlink, SITREPs on board)
 - Abort code: (if required) " _____"

3. Situation Update TEFACHR Format.
 - Threat: " _____"
 - General locations of surface-to-air threats not already covered.
 - Enemy Situation " _____"
 - General enemy disposition, avoid listing grids. Target grids will be addressed in CAS briefs.
 - Friendly Update: " _____"
 - General friendly situation and scheme of maneuver.
 - GFC intent.
 - Artillery: " _____"
 - List factor indirect fire assets.
 - This may include general direction of fire.
 - Clearance Authority: " _____"
 - Omit this if the speaker has control.
 - Define who has authority for brief, stack, mark, and control.
 - Hazards: " _____"
 - Towers, minimum safe altitude, weather, CBRN and explosives.
 - Remarks and Restrictions: " _____"
 - Ordnance restrictions, JTAC capabilities (e.g. laser, infrared)
 - Intent for aircraft (e.g. CAS, reconnaissance)
 - Other remarks

4. Gameplan or 5-line.
 - Do not transmit the numbers.
 - Units of measure are standard (metric) unless briefed.



- Lines 2, 3 and any restrictions are mandatory readbacks.
- The JTAC may request an additional readback.

5. 9-line (skip if 5-line)

- Do not transmit the numbers.
Units of measure are standard (metric) unless briefed.
- Lines 4, 6 and any restrictions are mandatory readbacks.
- The JTAC may request an additional readback.

6. Remarks/Restrictions

- Laser to target line (LTL) / pointer target line (PTL)
- Desired type and number of ordnance or weapons effects (if not previously coordinated).
- Surface-to-air threat, location, and type of SEAD.
- Additional remarks (e.g., gun-to-target line, weather, hazards, friendly marks).
- Additional calls requested.
- **Final attack headings or attack direction.**
- **Airspace coordination areas (ACAs).**
- **Danger close and initials (if applicable).**
- **Time over target (TOT) / time to target (TTT).**
- **Post launch abort restrictions (if applicable).**

7. Readbacks.

- 5-line: 2, 3 and restrictions mandatory readback.
- 9-line: 4, 6 and restrictions mandatory readback.
- JTAC may request an additional readback.

8. Target Correlation.

- Talk On:
Use plain english to guide the pilot to the correct target.
- Enhanced Target Description:
Provide a detailed description of the target and confirm that the pilot has the correct target in sight.
- Mark:
Apply the mark and confirm the pilot has correct target in sight.
- Video Downlink:
Use video downlink to confirm correct target.

9. Attack.

1. JTAC allow pilot to push to target.
2. Pilot confirms inbound heading.
3. JTAC clears the engagement.
(type-3 requires clearance right after step 8)
4. Pilot fires with appropriate brevity term for the weapon.

10. Assess Effects.

- Call out visual on splash.
- Assess damages and effect.

11. Battle Damage Assessment.

- Total damage.
- Any remaining threats (if required).
- Re-attack or continue to HA/HP?
 - If re-attack: request airframe to re-attack with any adjustments.
 - If continue to HA/HP: Provide routing and safety of flight back to HA/HP.

12. Routing / Safety of flight.

- Provide route out of combat area.
- Give safety of flight briefing (Step 1)

before beginning a new 9-Line.

Legend:

ASOC—air support operations center	JTAC—joint terminal attack controller
BDA—battle damage assessment	SALTR—size, activity, location, time, remarks
BOC—bomb on coordinate	TOT—time on target
CAS—close air support	TTT—time to target
FAC(A)—forward air controller (airborne)	VDL—video downlink
GFC—ground force commander	



Medevac or Casevac request

Table 82. 9-Line Evacuation Request		
Line	Item	Explanation
1	Location of pickup.	Grid coordinates using the military grid reference system (commonly referred to as MGRS) of the pickup site should be sent by secure communications. To prevent confusion, the grid zone letters are included in the message. Follow standard operating procedure for encryption of grid if using unsecure communications.
2	Radio frequency, call sign, and suffix (if used).	Frequency of the radio at the pickup site, not a relay frequency. The call sign of the person to be contacted at the pickup site may be transmitted in the clear.
3	Number of patients by precedence.	A—Urgent. B—Urgent-surgery. C—Priority. D—Routine. E—Convenience. If two or more categories must be reported in the same request, insert the word "BREAK" between each category.
4	Special equipment required.	A—None. B—Hoist. C—Extraction equipment. D—Ventilator.
5	Number of patients by type.	Litter + number of patients (Litter). Ambulatory + number of patients (Ambulatory).
6	(Wartime) Security of pickup site	N—No enemy in the area. P—Possible enemy in the area, use caution. E—Enemy in the area, approach with caution. X—Enemy in the area, armed escort required.
	(Peacetime) Number and type of wound, injury, or illness.	Specific information regarding patient wounds by type (ex: gunshot or shrapnel). Report serious bleeding, along with patient blood type if known.
7	Method of markings pickup site.	A—Panels. B—Pyrotechnic signal. C—Smoke (color). D—None. E—Other.
8	Patient nationality and status.	The number of patients in each category need not be transmitted. A—United States (US) military B—US citizen C—Non-US military D—Non-US citizen E—Enemy prisoner of war
	(Wartime) Contamination	C—Chemical. B—Biological.

9	(only if required).	R—Radiological. N—Nuclear.
	(Peacetime) Terrain description.	Describe obstacles and prominent features around the landing site (e.g., lake, tower).

Communication Brevity

These are the main terms used, this is not everything.

CAS	
Term	Meaning
Type 1	The JTAC can see both the Aircraft and the Target. (NO GPS MUNITIONS)
Type 2	The JTAC can see either the Aircraft and the Target.
Type 3	The JTAC can neither see the Aircraft or the Pilot or The aircraft is allowed to engage several times.
Danger Close	The target is within 400m creating a possibility that shrapnel hits friendly forces.
Broken Arrow	The target is on top of friendlies. CAS is being used as a last resort.
Fallen Angel	Aircraft crashed / had an emergency landing.
Bomb On Target (BOT)	The pilot will visually see the target and is able to identify the target.
Bomb On Coordinate (BOC)	The pilot does not have to see the target and will engage a coordinate instead.
A/C	Written term for aircraft.
Final Attack Heading (FAH)	Heading the A/C needs to maintain in order to receive clearance.
No Fire Area (NFA)	Specific area

Aircraft positioning	
Initial Point (IP)	Position from where the aircraft will start their run to the target.
Holding Area (HA)	Specified area where the aircraft will stay within.
Battle Position (BP)	Rotary Wing position for shooting while in a hover. Or a position for the start of the run to the target.
Return To Base (RTB)	-
Anchored	Aircraft is using a holding pattern around a point called the anchor.
Established	Aircraft is stable at the requested point or position.
Mean Sea Level (MSL)	Height measured above the sea
Angels	Height in 1000 feet (i.e. angels 5 = 5000 feet)
Cherubs	Height in 100 feet (i.e. cherubs 7 = 700 feet)
Ingress	Area the aircraft is coming in FROM
Egress	Area the aircraft is going out TOWARDS

Communication	
Oscar Mike	On the Move. (Not used on TADs)
Correlation	JTAC creates situational awareness for the pilot by discussing the target to ensure the pilot has the right target.
Keyhole	Method used to coordinate positions relative to the target area. (view Extra)
Interrogative	Question for the recipient.
No go	Will not comply
WILCO	Will Cooperate
HAVCO	Have Cooperated
CANTCO	Can not Cooperate
Shopping	Unofficial brevity for an airframe to request type 3 targets.
Jink	Directive to perform defensive maneuvers.

Laser Designator	
Laser on	Start/ acknowledge laser designation.
Lasing	The speaker is firing their laser
SHIFT (direction)	Shift laser energy. Note: Can be used to shift from the offset position onto the target. Also used during multi-aircraft attack to shift laser energy or target assignments.
SPOT	Acquisition of laser designation
CEASE LASER	Request to stop lasing.
DEAD EYE	Laser designator system inoperative.
NEGATIVE LASER	Laser energy has not been acquired.
10 seconds	Standby for laser ON call in approximately 10 seconds.
Laser to Target Line (LTL)	The direction in magnetic heading the laser is going towards. Allowing the A/C to identify the target side and the friendly side of the laser.
Self Lase	Request for pilot

IR Pointer	
Sparkle	Mark/markings target by infrared (IR) pointer. (JTAC marks the target with an IR pointer. Can be initiated by JTAC or aircrew. Proper aircrew response is CONTACT SPARKLE or NO JOY.)
Sparkle on	Response from JTAC (if initiated by pilot) that IR pointer is on.
Snake	Oscillate an IR pointer in a figure eight about a target. (Call made by exception for the JTAC to jiggle the IR beam on the target. This aids in distinguishing the friendly position from the target, verifies that the aircrew is looking at the proper IR pointer and can aid in the acquisition of the IR energy. Proper aircrew response is CONTACT SPARKLE, STEADY, or NO JOY.)
Pulse	Illuminate/illuminating a position with flashing IR energy. (JTAC uses pulse mode available on some IR pointers. Can be initiated by JTAC or aircrew. May be used by JTAC to emphasize that an enemy position is being illuminated by flashing IR energy, which is often used to identify friendly positions. Proper aircrew response is CONTACT SPARKLE, STEADY, or NO JOY.)
Rope	Circling an IR pointer around an aircraft to help the aircraft identify the friendly ground position.
Steady	Stop oscillation of IR pointer. (JTAC steadies the beam after a SNAKE or PULSE call. This can aid in verifying that the aircrew is looking at the proper IR pointer.)
Contact Sparkle	Acknowledges sighting of sparkle. Call acknowledging the sighting of a specified reference point (either visually or via sensor). After the SPARKLE call is made, the close air support aircraft should respond with NO JOY or SNAKE. Once the aircrew sees the IR energy and is able to discern between the friendly and target end of the pointer, a CONTACT SPARKLE call may be made.
Cease Sparkle	Discontinue sparkle activity.
Match Sparkle	Overlay requested target designator type. (Directive term for a second party to overlay an IR mark on an existing mark.)
Pointer to Target Line (PTL)	The direction in magnetic heading the pointer is going towards. Allowing the A/C to identify the target side and the friendly side of the pointer.

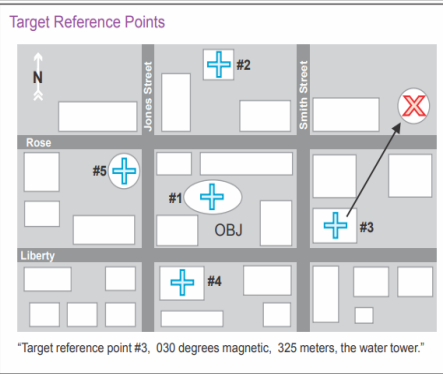
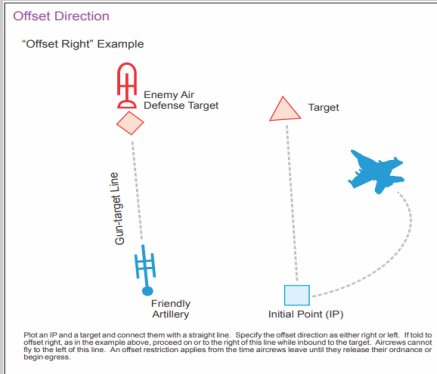
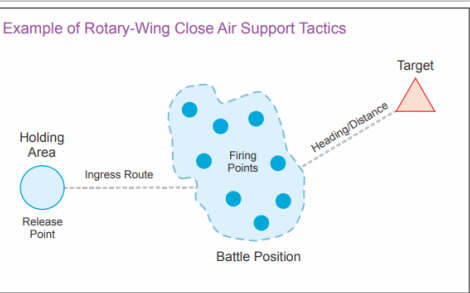
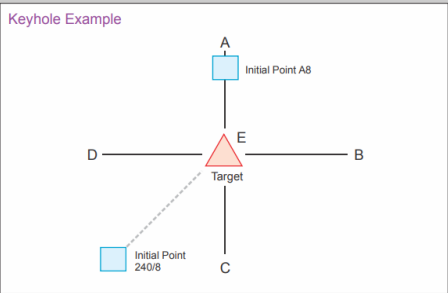
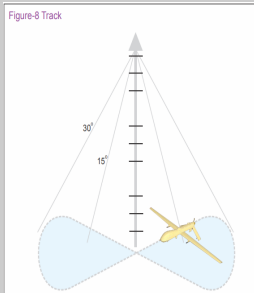
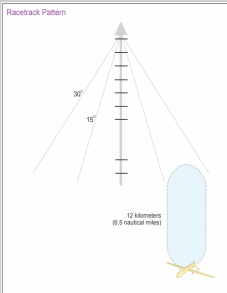
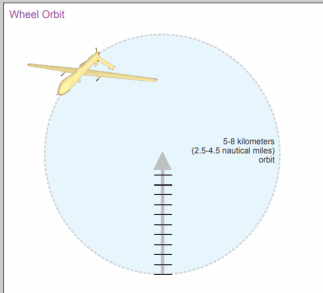
JTAC Clearance Terms	
Abort	Cease action/attack/event/mission
Cleared Hot	Type 1 and 2 control clearance to release ordnance on this pass.
Continue	Continue present maneuver, does not imply a change in clearance to engage or expend ordnance.
Continue Dry	
Cleared to Engage	Type 3 control clearance. Attack aircraft or flight may initiate attacks within the parameters imposed by the JTAC.
Set strike posture, say when ready to fire.	(Rotary Wing) Used for stand-off attacks. Pilot is instructed to ready his airframe to fire, and to call when ready to engage.
Push [DESTINATION]	Used to direct the aircraft to leave BP/HA and direct to destination. Can be used for directive to push target if no destination is given. EXAMPLE: "Push IP, call when established" OR "Push (target) when ready"
WARNING The word CLEARED will only be used when ordnance is actually to be delivered. This will minimize the chances of dropping ordnance on dry passes, further reducing the risk of friendly fire incidents. Nonstandard calls must be avoided at all times.	

Standard marking terms	
Blind	No visual contact with FRIENDLY aircraft/ground position. Opposite of VISUAL.
Visual	Sighting of a FRIENDLY aircraft or ground position. Opposite of BLIND.
Contact	Acknowledges sighting of a specified reference point (either visually or via sensor).
(target / object) Captured	Specified surface target or object has been acquired and is being tracked with an onboard sensor.
Looking	Aircrew does not have the ground object, reference point, or target in sight. Opposite of CONTACT.
Tally	Sighting of a target, nonfriendly aircraft, or enemy position. Opposite of NO JOY.
No joy	Aircrew does not have visual contact with the TARGET. Opposite of TALLY.

Weapon terms	
Guns	Miniguns, 30mm, etc.
Rifle	AGM, Hydra 70
Ripple	Aircraft releasing multiple weapons
Pickle	Dumb / GPS guided bombs, GBU-53, GBU-37
Paveway	Laser guided bomb, GBU-12
Splash	JTAC spotted ordnance hitting the ground.
Shack	Pilot spotted ordnance hitting the ground.
Winchester	Airframe out of ordnance

	Gun-to-Target-Line (GTL)	Any ground artillery that is a possible factor for the aircraft.
	Jink	Directive to perform defensive maneuvers

JTAC Extra



General Visual Perspectives for Talk-On Consideration

	Ground	Fixed-Wing	Rotary-Wing	Unmanned Aircraft System
Vertical Relief	Good	Poor Note 3	Good	Poor Notes 2, 3
Horizontal Relief	Poor	Good	Poor Note 3	Good
Color Distinction	Good Note 2	Poor* Notes 3, 4	Good Note 3	Good Notes 1, 2, 3
Communications Delay	Not Significant	Not Significant	Not Significant	Significant

This figure assumes the environment is sky clear, visibility unrestricted.

* Aircraft greater than 15,000 feet slant range from the target will likely be unable to discern color with the naked eye. The use of binoculars or other sensors essentially "decreases" slant range to a target to a distance at which the color may be determined.

NOTES:

- (1) Sensor dependent
- (2) Slant range dependent
- (3) Altitude dependent
- (4) Enhanced when aided (binoculars)

